

Descriptive Statistics and Technical Interpretation

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Introduction

This report presents a technical study of descriptive statistics as applied to measurement data from the Biochemistry, Electronics, and Mechanical domains. It covers measures of center and spread, distribution shape, sample size, outlier handling, plot selection, correlation versus causation, and domain-aware comparability, with the aim of explaining what summary statistics reveal, what they conceal, and when a conclusion drawn from them is valid.

I. Descriptive Statistics and Why They Matter

Descriptive statistics refer to the summary of a dataset in a clear and concise way.

They are used before cleaning, visualization, and other forms of data manipulation because they give a clear overview of the entire raw dataset before any changes are made to it — so the analyst understands what the data looks like before altering it.

Statistics summarize data but cannot explain a pattern, because a summary number only describes what the data looks like — it carries no information about the underlying process or mechanism that produced it. Two completely different processes can produce the exact same mean.

Example: the average package of a company's employees. The average could be high enough to place the company among the top tiers, but in reality the dataset had a large number of outliers that poorly explained the actual pattern of the data.

II. Center and Spread

Mean refers to the overall average of a variable, calculated by summing all its values and dividing by the number of values. Median refers to the middle value of a variable when all its values are arranged in order. Minimum and maximum refer to the lowest and highest numerical values of a variable, respectively. Range refers to the difference between the maximum and minimum values of a variable, giving a measure of how spread out the data is. Variance refers to the average squared differences from the mean, and is used to measure how spread out the values are using the mean as reference. Standard deviation is the square root of variance, and is used to measure the typical distance between each point and the mean. IQR (Interquartile Range) is the distance between the 25th percentile (Q1) and the 75th percentile (Q3), and is used to measure the spread of the middle 50% of the dataset.

The center represents a typical or middle value, while the spread describes the variability, or how far apart the data points are.

The mean becomes misleading when outliers are present in the dataset, since even one or two extreme values can pull the mean away from what most of the data actually looks like, giving a misleading picture of the "typical" value.

The median is more stable against extreme values because it depends on position, not magnitude — it only cares about which value sits in the middle rank. An extreme value can shift the median by at most a few positions, but it cannot pull the number itself the way it pulls the mean.

High standard deviation in a dataset means that each data point is, on average, far away from the mean.

IQR is useful when data is skewed or has outliers because it presents the spread of the middle 50% of the dataset, deliberately excluding the top and bottom 25% — which is exactly where outliers and skew tend to live. This means IQR is not dragged around by extreme values the way the mean, range, or standard deviation are, giving a clearer idea of the dataset's typical spread.

III. Distribution Shape and Sample Size

A distribution gives a much fuller picture of a numerical variable than a single summary statistic — it shows where values cluster, how skewed or symmetric they are, and where outliers, if present, sit within the overall spread of the data.

Skewness indicates that the data is asymmetric — more data is present on one side of the graph. Clusters indicate there might be sub-groups within the raw dataset. Gaps indicate either a limit or that no data was found in that range. Spread indicates the overall variability of the data.

A statistic from five readings is weaker than one from five hundred because there is not much that can be derived from just five readings — outliers can severely impact the analysis. With five hundred readings, individual noisy or extreme points get diluted out by the sheer volume of data, so the resulting statistic converges much closer to the true underlying value.

Sample size matters before trusting any central tendency or visual pattern for the same reason: outliers can severely impact a smaller dataset and interfere with the actual analysis. Larger sample sizes are more resistant to noise and give a much better analysis of the data.

Example: testing three resistors from a production batch and finding their average resistance noticeably higher than the datasheet value could look like a defective batch. But three samples are not enough to distinguish a genuine manufacturing defect from ordinary batch-to-batch variation — with only three readings, one slightly off resistor swings the average significantly, and there is no way to tell if it is a fluke or a real pattern without testing more units.

IV. Outliers and Domain Context

An outlier is a data point that is further away from the central part of the dataset.

Not every outlier is automatically an error, because it is natural to have extreme values that are still plausible depending on the variable — for example, someone scoring 100 on a difficult test where the average is 50.

An outlier should be investigated instead of removed when there is a finite possibility that such a value is plausible — for example, a resistor reading 100 ohms when the average is 30 ohms. In that case, the value should be checked against the actual measurement conditions rather than assumed to be correct or wrong outright. A value that has no chance of being legitimately possible for a variable, on the other hand, can be removed — for example, a pH reading above 14 when the average is around 8.3, since pH cannot physically exceed 14.

A. Biochemistry

A pH reading of 2.1 in a buffer solution expected to sit around pH 7 could indicate contamination, an incorrectly prepared buffer, or a miscalibrated probe.

B. Electronics

A sudden current spike during otherwise steady-state testing could indicate a genuine overload condition, a short circuit, or a fault in the measurement setup itself.

C. Mechanical

An unusually high force reading during a tensile test could mean the material is genuinely stronger than expected, a unit mismatch between N and kN, sensor saturation, or a rare but real shock event captured by the load cell.

V. Plot Selection and Interpretation

A histogram is used to display the distribution of a single numeric variable, split into many bins, and is used to observe the spread of data. Example: plotting hundreds of pH readings across batches to see whether they cluster tightly around a set point or are widely scattered.

A box plot is used to compare groups side by side, and can be used to measure the spread of a numeric variable across multiple categories. Example: comparing tensile strength across three different alloy batches on a single chart.

A bar chart is used to display a single numeric value across categorical groups. Example: comparing average voltage output across five different circuit boards.

A scatter plot is used to show the relationship between two continuous variables, and can be used to see the kind of relationship and whether they move together. Example: plotting force against displacement in a tensile test to check whether the relationship is linear.

A line plot is similar to a scatter plot, but connects points in order (sequence or time). Example: voltage measured over time during a battery discharge test.

Title, axis labels, and units matter a great deal in data visualization, as they give users a clearer understanding of the data being presented. They also provide the right units for measurement and make it easier to derive a relationship between two or more variables. Without them, users would not know what variables have actually been compared.

VI. Correlation, Causation, and Confounding

Correlation refers to a relationship between two or more variables that tend to move together — either increasing or decreasing in tandem. Causation refers to the phenomenon wherein a change to one variable directly produces a change in another variable, i.e., changes to X cause changes in Y.

Two variables rising together does not necessarily prove one caused the other, because it can be due to other reasons: a third variable might be the actual cause behind both, or it could simply be a coincidence. Two variables rising together alone is not sufficient proof of causation.

A confounding variable refers to an existing hidden variable that influences both variables being compared — making them appear related to each other, when the real driver is actually the confounder rather than either variable itself.

Example: hours studied and exam score often rise together, which could be misread as "studying more directly causes a higher score." But a confounder such as overall interest or discipline in the subject could be driving both — a more engaged student naturally studies more and prepares more effectively, so study time itself is not necessarily what produces the higher score.

VII. Domain Measurement Interpretation and Comparability

An overall mean across pH, voltage, current, force, and pressure is meaningless because these are completely different physical quantities, measured in different units, with different domains and effects. Averaging them together is mathematically meaningless — similar to averaging a temperature with a distance — since the resulting number does not correspond to anything real.

Units play a vital role in providing meaning to a numeric value, as they determine the property against which the value was measured.

Values should be grouped by domain, measurement type, unit, and condition before interpretation, because only then will the appropriate values be grouped correctly, which can then be analysed and visualised properly — giving a much clearer and more reliable idea of the actual relationships and patterns in the data.

Valid Comparisons

- Average voltage under low-load versus high-load conditions, measured on the same circuit.
- Average tensile strength across different alloy samples, tested on the same rig under the same conditions.
- Average pH before and after adding a buffer, within the same reaction.

Invalid or Misleading Comparisons

- Comparing a pH value numerically against a voltage reading.
- Averaging force readings where some are recorded in newtons and others in kilonewtons, without converting them to the same unit first.
- Comparing current readings from two different circuits tested under different load conditions, and treating the difference as meaningful.

Extra information that should be requested from domain teams before analysis includes whether the standard units used for each variable are clearly mentioned, what data cleaning or preprocessing has already been applied, whether the data is raw or processed, and the specific condition under which each measurement was recorded.

VIII. Final Reflection

Statistics can be arithmetically correct while still being scientifically or practically misleading. A value can be technically valid but meaningless if its unit is missing, since a number alone carries no real information without knowing what it is measuring. Mixing different measurement types — such as averaging pH with voltage — produces a number that is mathematically computable but does not correspond to anything real, since the quantities being combined are not comparable to begin with. A statistic drawn from too small a sample can look precise while actually being unstable, since a single extreme reading can swing it significantly with no way to tell if it reflects the true process. Outliers, similarly, can pull a correctly calculated mean far from what the data actually looks like, and dropping or keeping them without investigation can distort the analysis either way. A comparison can be numerically sound but still misleading if the context behind it — the condition, calibration, or unit — is not known. Correlation is often reported as though it proves causation, when a hidden confounding variable could be driving both. Grouping the wrong values together, ignoring domain, unit, or condition, can make even a correct calculation say something false about the world.

IX. Review Questions

- 1) If mean and median are far apart, it means that there are outliers or extreme values in the data, and they are pulling the mean away from the median.
- 2) Standard deviation alone is not enough because it just gives the typical distance between each point and the mean — without the unit, that distance is meaningless, and without the experimental condition, it is unclear whether the spread is normal noise or a sign of an unstable process.
- 3) Not every outlier is wrong — some outliers show the plausibility of a value, indicating that a certain event is genuinely possible. For example, someone getting a perfect score on a paper where the average is 45.

- 4) Attendance is a valid data type because 105 is numerically well-formed, but attendance percentage can only lie between 0 and 100 percent, so 105 percent falls outside the allowable range for that variable, rendering it an invalid value.
- 5) An average of pH, voltage, current, and force is meaningless because they are completely different physical quantities measured in different units — the resulting number does not correspond to anything real.
- 6) The median is more robust to outliers than the mean, because it depends on position, not magnitude, while the mean is pulled directly by the value of every point including extreme ones.
- 7) A bar chart only shows a single summary numeric value against every category, whereas a box plot shows the median, Q1/Q3, and the spread via IQR, along with individual outlier points.
- 8) A positive correlation between attendance and score means the two variables tend to move together; it cannot be concluded that attendance causes the higher score, since a confounding variable such as discipline or interest could be driving both.
- 9) "The average of pH, voltage, current, and force across the dataset is 4.3" is arithmetically correct but scientifically useless, since these quantities cannot be meaningfully combined.
- 10) Extra metadata worth requesting includes the standard unit used for each variable, whether the data is raw or processed, what cleaning steps (if any) were applied, and the exact condition under which each measurement was recorded.

References

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